



Faclon Labs Data and AI Infrastructure

Decision Intelligence at scale to unlock Latent Potential of each Plant

Improve Production
Throughput and Yield

Reduce Operational
Cost and Wastage

AI Agents Enable Decision
Intelligence at Scale

www.faclon.com



Faclon Labs at a Glance

Leaders in Industrial Data Integration, Visualization, and AI based Optimization



50+
Customers

200+
Deployments

60k+
Assets
Connected

5B+
Data Points
Daily

120+
Employees



Rishi Sharma
CEO
B.Tech IIT Bombay



Utkarsh Srivastava
CTO
Ex-White Hat Hacker



Archit Naraniwal
Chief Sales Officer
B. Tech IIT Bombay



Parth Chaudhary
President
B. Tech IIT Bombay



Vivek Singh
Chief Data Scientist
B.Tech IIT Kanpur



Parag Patil
Chief of Strategic Relations
Ex Wipro

Backed By



Paula Mariwala
Founder, Stanford Angels



Sarjan Shah
MD, Group Satellite



Jay Kotak
Co-head, Kotak 811



Siddharth Mehra
ED, Savita Oil

Key Customers

Trusted and Ideal digital partner for Industrial setups, commercial setups and OEMs

Automobile & ancillary



TATA HITACHI



Consumer Goods



Metals



Cement



Pharma



Textile and Consumer Durables



LUMINOUS



Commercial Infra



Energy, Oil & Gas



OEM tie-ups



Partnerships



What we do, How we Deliver

Powerful Ecosystem of platform offerings for rapid digital transformation

Critical Outcomes



Automating Decisions

- Diagnostic Analytics
- Prescriptive Analytics
- Process Control
- Digital Twin Simulations



Problem Visibility

- KPIs Benchmarking
- Dashboards, Alerts & Reports
- AI based Detection Models
- Golden Batch Analytics



Data Unification

- Integration with all IT and OT data sources
- Workflow Digitization
- System of Records
- Data Contextualization



The Operating System for Physical Infrastructure

Hyper Customised Factory Specific Intelligence at Scale

01 Production
Throughput & Quality

03 Maintenance
Availability & Maint. Cost

02 Energy & Utilities
Equip. Efficiency & Specific Consumption

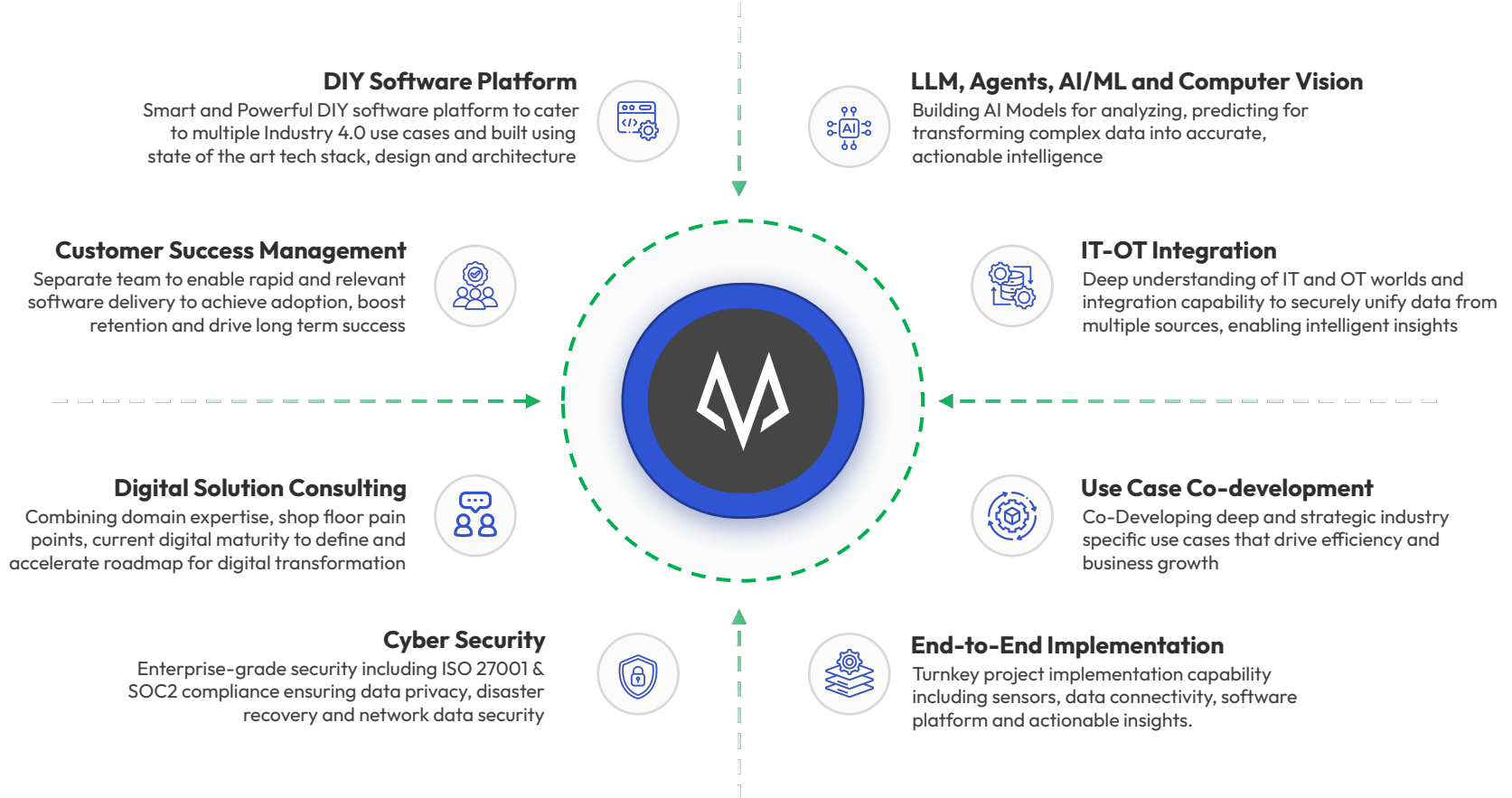
04 Safety
Compliance & Incidents



Connecting People, Assets, Process and Data at Scale

Ideal Mix of Capabilities to Achieve Transformational Results

Ideal digital transformation product to measure, find issues and improve productivity



Platform Architecture Overview

Powerful Ecosystem of platform offerings for rapid digital transformation



CxO Suite | Digital COE | Plant Heads | Plant Managers

I/O Connect

IT

ERP, CRM, MES, Databases

OT

SCADA, DCS, PLC, Controllers, LIMS

IoT

Sensors, Gateways, MQTT

Media

Camera

Logs

PDF, Excel, Log Sheets

Connect Everything

I/O Sense

Data & Context

Unified Data Fabric

MuSTO Architecture for unified data access across multiple storage types

System of Records, Data Lake

Unified Namespace, MCP

Enterprise Context Layer for AI agents

Capabilities

Business Intelligence

Analytics

Trigger Workflows

App Libraries

Single Source of Truth

Deepsense

AI / ML Ops

Fine Tuned Models for prediction, optimization, simulations, forecasting.

Computer Vision

Next Generation Computer Vision and video analytics

Bruce AI

Vibe Engineering

No-code prompt based tool to develop dashboards and features with AI

Chatbots and AI Agents

Build next generation Agentic AI to automated decision making and workflows

Activated with AI

Digital Use Cases

One-platform, unlimited applications

Production



OEE &
Downtime
Monitoring



IT-OT Integration
and Data Lake



Digital Logbook
& Workflow
Automation



Automated
Process control



Digital Twin



Product
Traceability



LIMS & Quality
Management



Vision - Surface
Defect
Detection



Soft Sensor



Golden Batch
Analysis

Energy & Utilities



Energy
Management



Utilities
Efficiency
Management



Water
Management



Intelligent HVAC
Automation



Solar and Wind
Monitoring



AI Agent for
Energy Audit
and RCA



Soot Blower
Automation



Power Plant AI
for NHR
Reduction

Safety



Fire Hydrant
Monitoring



Vision - PPE &
Road Safety

Maintenance



Steam Trap
Failure Detection



Vibration
Monitoring
(CBM)



VFD Health
Monitoring



Busbar / Bus Duct
Temperature
Monitoring



Vision - Foreign
Particles



UPS Health
Monitoring



Advance
Maintenance
Management



Vision - Conveyor
Defect Detection



Vision - Particle
Size Analysis



Relay &
Transformer
Monitoring



Power Quality
Monitoring



Anomaly
Detection

	Model Digital Plant	- Deployed a flagship Model Digital Plant for JSW Cement's integrated plant at Nandyal, integrating DCS systems, energy meters, vibration sensors, solar inverters, ERP, and lab equipment to create a single source of truth — a unified data lake with tailor-made dashboards for every department. - Implemented AI use cases including cement strength prediction, C3S prediction, cyclone health prediction, and kiln fuel-ratio optimization to increase AFR, along with root-cause analysis and recommendations for high specific energy consumption in mills and kilns.
	Energy Management, Power Plant AI and Reliability	- Digitized 2,500+ energy meters across 4 plants — Vilayat, Kharach, Harihar, and Indonesia — with deviation analysis at every level of energy distribution. Prevented critical motor failures through motor-load-based alerts. The complete platform is deployed on Grasim's private cloud, catering to all the use cases from a single environment. - Implemented online steam trap failure detection for 250 HP steam traps across 2 plants, 100+ vibration sensors on critical motors and gearboxes to cut unplanned downtime and maintenance costs, and health monitoring of 200+ VFD drives and 100+ bus bars to extend asset life and reduce breakdowns. - Implementing AI-based solutions to reduce NHR for 175 MW captive power plant comprising 5 boilers and 5 turbines.
	Cell House Performance	Implemented an integrated cell house monitoring system across 66 cells, by implementing load cell sensors, wireless cell temperature sensors and integrating existing crane PLC and cell current data to map stripping activity, proactively manage cell temperature and monitor cell efficiency in real time. Scale up underway for 330 cells across 3 plants.
	Vision AI for Conveyor Health	Monitoring the health of three conveyor systems using vision cameras — shuttle conveyor (proximity switch fault detection), tray conveyor (alignment, wheel wobbling, and misalignment), and belt conveyor (spillage, wear and tear).
	Power Plant AI	Implementing an AI-based solution for a 91 MW power plant to reduce auxiliary power consumption and net heat rate (NHR) at Zawar Plant. Integrating with their historian (DCS), EMS, Lab data. Project aimed for 1-percentage point reduction in APC.
	Power Balance	Implemented a Platform to balance power between generation - 6 captive power plants, Grid (Re-PPA, Discom), and consumption - 8 consuming plants and excess generation on IEX. The system automate day ahead and intra-day 15-minute power mix scheduling to prioritize the lowest-cost mix across 550 MW captive thermal generation, the 180 MWp solar PPA, interplant transfers, and IEX while enforcing NRLDC/regulatory limits, CPP minimum PLF, and transfer constraints.

	IT-OT Integration	- At Welspun's LSAW unit in Anjar, data from 18 data sources — PLCs, SCADA systems, SQL databases, and Excel files — covering ~4,000 tags across critical stations such as JCO, UT, Hydrotester, and Welding is consolidated into a central server, creating a unified environment for analysis and reporting. - Management and plant teams use KPI dashboards covering station-wise cycle time, energy consumption, pipe production, and machine utilization for real-time visibility — enabling faster response to downtime, reduced production losses, and improved plant productivity, energy efficiency, and quality. Over five months, analysis of 14,777 pipes demonstrated a 10.2% reduction in cycle time and a 5.0% improvement in energy intensity.
	OEE and Traceability	Integrating 400+ machine PLCs — including CNCs, extruders, braidiers, and testing machines — across four units in Chennai, with real-time tracking of key KPIs such as OEE, availability, performance, quality, production, downtime reasons, line speed, thickness, Cpk/Ppk, and process parameters. Also enabled end-to-end traceability by mapping every shop floor machine run to its order code.
	Power Plant Automated RCA	Developed AI-agents for a 21 MW cogeneration plant that leverages plant DCS and Lab data to identify problems / deviations and provide root cause and recommendations. It covers all possible problems related to maintenance and process control for boiler, turbine, generator and aux equipments. The solution is developed in collaboration with a domain SME with 40+ years of experience.
	APC for Coal Feeding	Implementing automated process control for 7 coal feeders of 21 MW power plant with AFBC boiler to reduce process variability and maintain bed stability.
	AI based Soot Blower Automation	Delivered closed loop AI based sootblower automation for 16 MW recovery boiler having 29 soot blowers. Saved 20 TPD aux steam worth ₹1 cr in annual savings. The solution helps reduce wash cycle, reduce tube erosion, improve flue gas heat recovery.
	RDH Optimization	- Implemented Heat Recovery Optimization basis accumulator tank temperature & level management during black liquor displacement, and controlled MP steam and recirculation pump parameters - reducing cycle time and steam consumption while maintaining cook uniformity. - Implementing digester and cooking optimization analytics for the RDH batch fibre line at SPB Erode, integrating DCS data, black liquor concentration, fresh alkali charge and strength, and wood species/quality to build live soft sensors that predict kappa number at pump-out and also predict post-washer brightness - enabling proactive cook-endpoint control.
	Predictive Maintenance	Deployed 1,000+ vibration sensors across five JSW Energy plants, detecting 15+ critical failures at an early stage. Reduced unplanned downtime, revenue loss, and maintenance costs — delivering ₹1.3 Cr in savings over the last two years, validated by their finance team.

	Steam Trap Monitoring	Implementation real-time health monitoring of 800+ steam traps at the Bathinda site, using a fully wireless, battery powered, temperature-based solution to reduce MTTR to less than day, reduce steam losses & improve reliability.
	HVAC, Steam trap & Solar Monitoring	- Implemented a fully automated IoT-based system for chiller plants, AHUs, and packaged units at two locations, - Implemented real-time steam trap failure detection for 100 HP and LP steam traps. - Deployed centralized solar efficiency monitoring across 43 sites to track energy performance.
	LIMS Platform	Implemented a comprehensive, integrated LIMS platform spanning 10+ plant sections — RMH yards, Coke Oven, Blast Furnace, and Steel Melting Shop (SMS) — enabling real-time monitoring, process adjustments, capability analytics, traceability, faster root-cause analysis, and automated reporting and alerts. The system is integrated with lab machines, ERP, and MES.
	Vision AI for Safety	Implemented IO Vision across 196 cameras to enforce plant safety compliance with ISO 39001:2012 (Road Traffic Safety Management) and ISO 45001 (Occupational Health & Safety Management) standards, covering 5 use cases including PPE and hard hat detection, fall and loitering detection, entry–exit monitoring, intrusion detection, and speed monitoring.
	IT-OT Integration	Implemented on-premise centralized monitoring by integrating all machine DCS systems and energy meters across the plant, enabling real-time monitoring, data historian, reporting, and alerting.
	Energy and Utilities Management	Deployed a Utility Management System for ITC at Trichy, Uluberia, and Mysore, digitizing 600+ assets covering energy meters, BTU meters, compressed air flow meters, water meters, and HSD and LPG systems. Scale-up to seven more plants is underway.
	Energy Management	Deployed power analytics and energy management covering 700+ energy meters across 3 plants.
	Bombay Airport Digitization	- Monitoring and controlling 75 AHUs at the domestic terminal to maintain temperature and improve energy efficiency. - Deployed wireless, battery-powered ambient temperature and humidity sensors across 300 locations to ensure consistent, comfortable cooling throughout the terminal. - Installed health monitoring for 200 UPS units and temperature monitoring for 70 bus bars to ensure power backup reliability and proactive issue detection.
	Adani Natural Resources	Implemented a pilot for real-time monitoring of exploration drill rig performance for Adani Natural Resources.

	Cold Storage Monitoring	Implemented cold storage temperature monitoring across Flipkart's 1,000 dark store locations in India.
	Smart Maintenance	Delivered CMMS-SAP integrated maintenance workflows & Installed partial discharge & vibration sensors, enabling automated work orders, technician assignment, material planning, PR and reservation creation, PO lifecycle tracking, man-hour logging, and real-time closure for both preventive and corrective maintenance.
TATA HITACHI	Process Excellence	Digitized end-to-end boom manufacturing workflow at the Kharagpur plant, replacing manual check sheets with real-time production, quality, downtime, and operator productivity tracking across fabrication operations. - Established complete digital traceability with first-pass-yield monitoring, failure analytics, and ISO-compliant document control, enabling faster root-cause analysis and data-driven shop-floor decisions.
	Discrete OEE	- Deployed OEE & Production Planning module across the bottle manufacturing line (7 Stations), integrating relay-based production signals, buzzer alerts for production stoppages, Andon-based downtime classification into Production/Maintenance/Quality losses. - Enabled SKU-wise production and energy analytics through direct integration with energy meters, covering production, energy consumption, SEC, and historical performance-based ideal cycle time recommendations for improved planning and benchmarking.
	Process Excellence	Integrated 14 environmental chamber machines with real-time monitoring of temperature, humidity, and pressure across product testing workflows. Digitized end-to-end test planning, execution, and approval workflows with multi-level validation and operator-driven test logging. Enabled energy-optimized test scheduling and ERRC-based recommendations using historical test data to improve utilization and reduce defects.
MAHLE	SOP Adherence	Enforced computer-vision-based SOP adherence with PLC interlocks on a 15-step carburetor assembly line — enabling real-time step validation, preventing sequence deviations, and reducing rejections by ~50%.
	OEE and Cycle time Tracking	Transformed 40+ injection moulding machines at Godrej's vendor sites with real-time OEE, downtime, and cycle time analytics, plus a mould recommendation engine.
LUMINOUS	Process Excellence	Integrated with Paste Mixer PLCs to capture key process parameters and batch cycle times, along with a batch quality input module for golden batch analytics, enabling comparison of batches to identify deviations, bottlenecks, and drivers of process quality.
	Quality Traceability	Implemented an end-to-end digital thread for battery testing — mapping QR-scanned SKUs to HRD recipes and SQL databases to ensure fail-safe label printing only for quality-passed units.

Customer Cases (On going)

	IT-OT Integration	IT-OT Integration for Color Coating Lines across Khopoli & Rajpura plant, integrating 8,000+ tags with real-time monitoring and visualization of critical operational and performance KPIs.
	Digital Twin	- Digital twin for DRI & Pellet Plant: Deployed a kiln AI model and tag-driven analytics across MSP Steel's sponge iron division – converting DCS, lab, and asset-health data into live stability, efficiency, and quality indices on one common backbone. - The payoff: A per-kiln decision engine that guides operators (Increase / Maintain / Reduce) plus predictive alerts for accretion risk and asset health → longer campaign life, fewer unplanned stoppages, optimized fuel rate.
	Thermal Profiling for Copper Cell House	- Thermal camera based real-time cell house temperature profile monitoring during electrolysis to detect early signs of short circuits ("shots") by monitoring temperature rise and enable timely intervention. - Benefits : Higher efficiency, cutting electricity losses, minimizing process disruptions, and sustaining consistent copper deposition efficiency on the cathode.
	Anode Current Distribution	Implementing Anode Current Distribution(ACD) monitoring pilot for 2 pots having 80 anodes. The solution involves a customized hardware every anode to measure current in real time. The data enables early detection of localized anode effect (AE), anode level - resistance & current imbalance to provide operational insights to stabilize pot voltage and reduce energy waste.
	Quality Prediction	Implementing AI model that analyses polymerization process batch data, DCS data, MES data, Lab data to provide realtime diagnosis into batch cycle time, and predicting in-process Viscosity and % Solid Prediction. Objective to cut cycle time, improve yield and quality, reduce rework.
	Renewable Monitoring	Monitoring and performance optimization of 160+ MW across 28 sites including solar, wind, and hybrid renewable energy assets, including plant performance, generation losses, and operational KPIs.
	Quality Prediction	Implementing batch process optimization analytics for the Profenofos manufacturing line at Sumitomo Chemical, Bhavnagar, integrating DCS and lab data to build a predictive quality soft sensor for BCP, DTCL, and triester purity – eliminating the 1-2 hr lab turnaround and enabling live cook-extension and reprocessing decisions to cut cycle time and improve yield & quality.

We're here as your

Digital Backbone to Drive Operational Efficiency and Decision Intelligence

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